Digital Image Processing

HW 3: Adaptive Median Filtering

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11 October 2025

Abstract

In many fields, the analysis of images can be crucial for solving a problem, such as in the medical field with x-rays and MRIs. Oftentimes, these images can come with a noise that disrupts the analysis of the image, and an appropriate method needs to be used to eliminate the noise without losing the important details of the image. In this report, the adaptive median filter will be used to restore very noisy images, while preserving the details and edges.

Introduction

Given a few samples of images ranging in noise level, a variety of filters will be applied to them to test how well they compare to their original image. A way to test similarity of two images is by using the mean square error, which takes the average squared difference of each of the pixels in the images. The value obtained from this is considered the error, and the higher the error is, the least similar the images are mathematically. This report will touch on the comparison of the output images using these filters and the MSE values calculated from them. The different filters that will be used in this experiment are the gaussian filter, median filter, and the adaptive median filter.

Experiments and Results

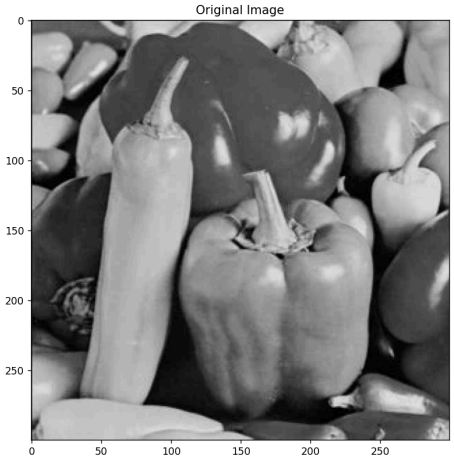


Figure 1. Original image of vegetables used in both experiments

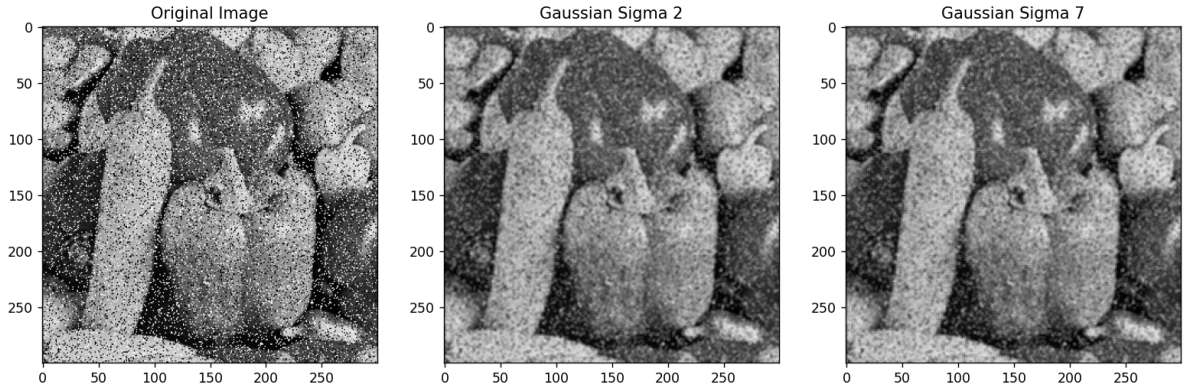


Figure 2. Noisy image with gaussian filters of sigma 2 and 7 applied.

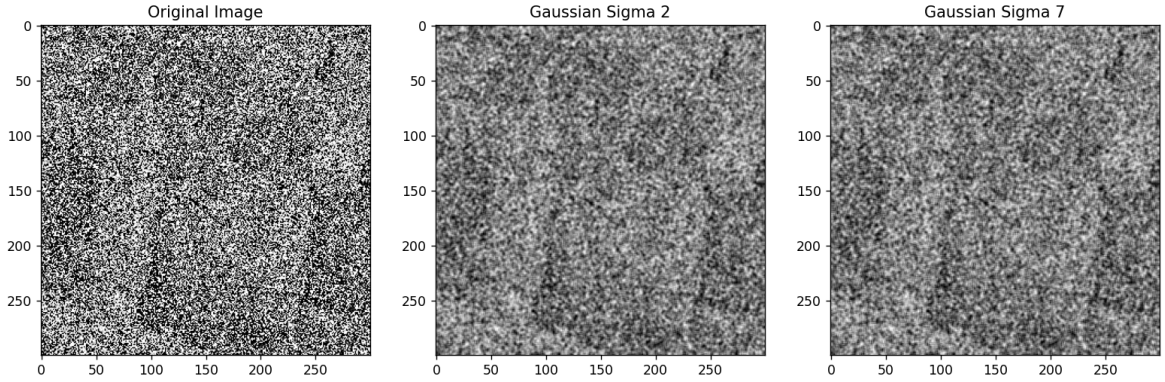


Figure 3. Noisy image with gaussian filters of sigma 2 and 7 applied.

As seen from figures 2 and 3, the gaussian filter doesn't achieve very successful results in restoring the image, since this filter uses the average value of each neighborhood to alter the image, it ends up blurring the image a bit and smoothing out the needed edges. This is not ideal when trying to restore an image.

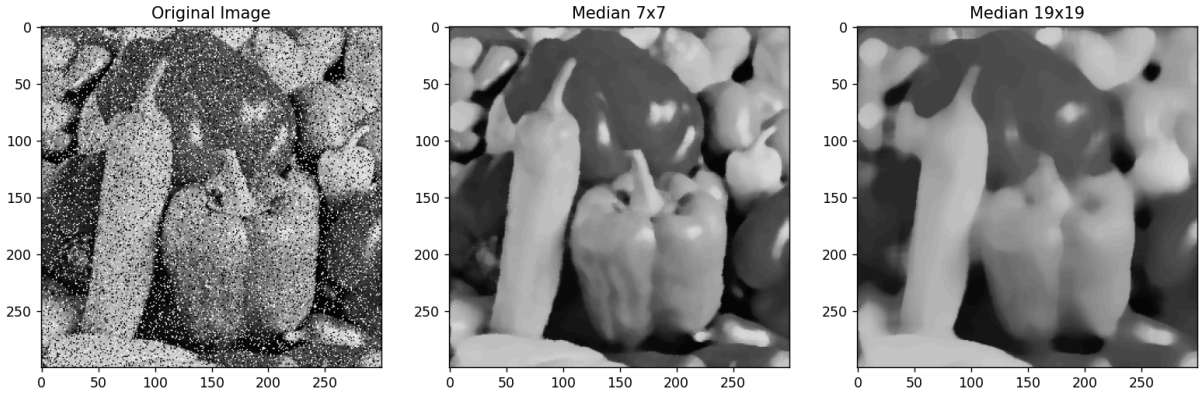


Figure 4. Noisy image with 2x2 and 7x7 median filters applied.

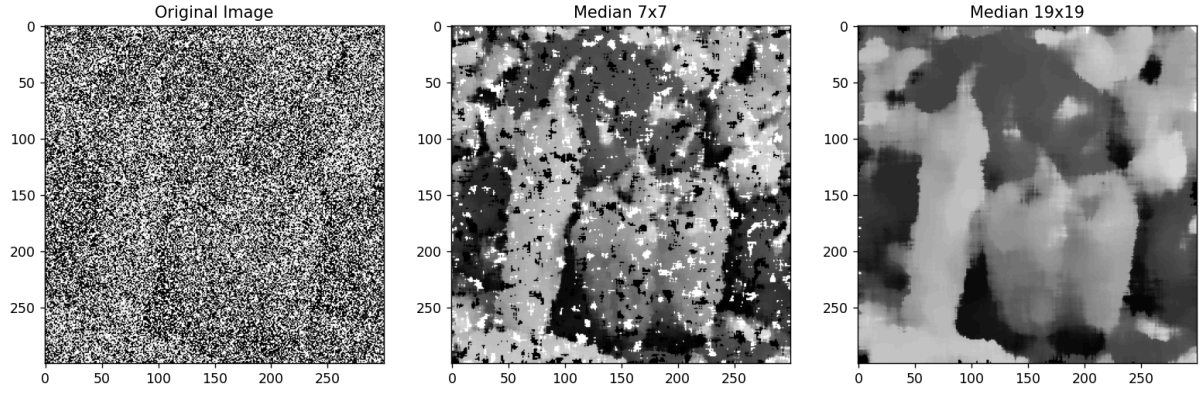


Figure 5. Noisy image with 2x2 and 7x7 median filters applied.

The median filter had a much better performance in removing the noise of the image. This is due to the fact it replaces the pixel values with the median value of the neighborhood,

which allows for the salt and pepper noise (outliers) to be removed. In figure 4, it shows how this filter at high noise and neighborhood sizes can produce artifacts, causing more distortion. These results still show a loss in detail, seen in some of the highlights and shadows of the image.

MSE (error)	Noisy	Gaussian Filter (sigma=2)	Gaussian Filter (sigma=7)	Median Filter (7x7)	Median Filter (19x19)
Test1Noise1	22.17	88.46	89.71	27.41	52.16
Test1Noise2	88.93	105.65	105.79	54.98	58.83

Figure 6. Table of the MSE values for each test image and the given filter applied to it.

According to the graph in figure 6, the MSE error reports to get worse after the filters are applied to the noisy image. I believe that this is occurring because MSE compares the exact pixel values from each image, which doesn't take into account whether the images actually look more similar or not. The filters are modifying the original image further, therefore changing the pixel values more and increasing the MSE.

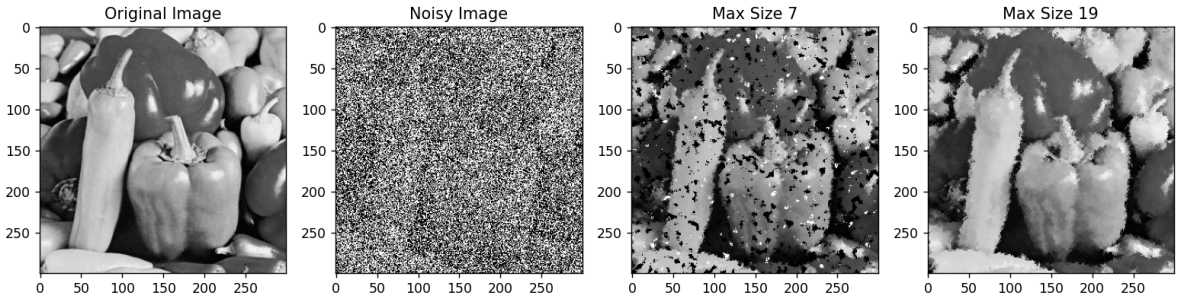


Figure 7. Noisy image with max window size 7x7 and 19x19 adaptive median filter applied.

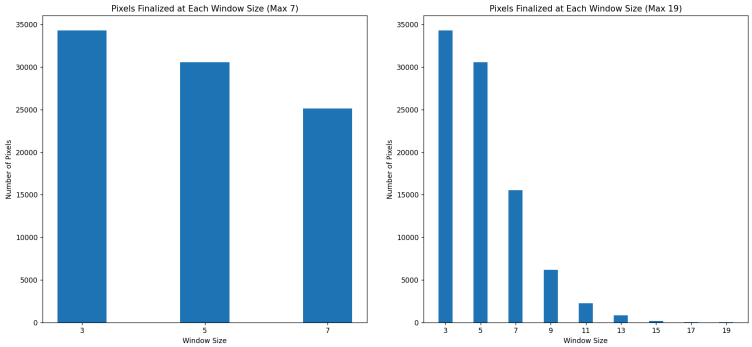


Figure 8. Graph of the window sizes in which the pixels of figure 7 were finalized in using the adaptive median filter.

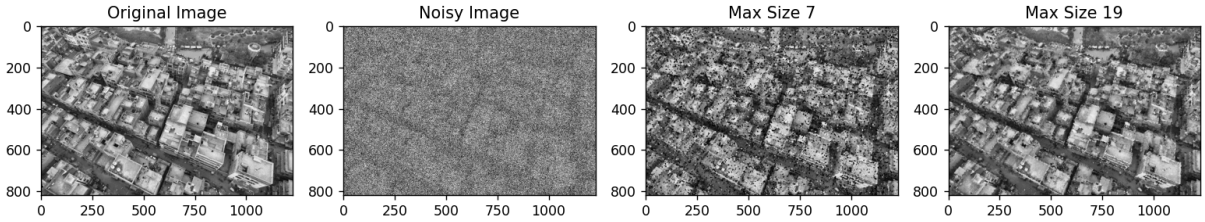


Figure 9. Noisy image with max window size 7x7 and 19x19 adaptive median filter applied.

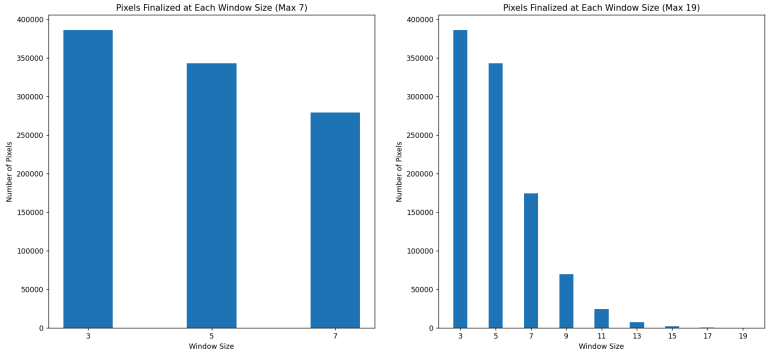


Figure 10. Graph of the window sizes in which the pixels of figure 9 were finalized in using the adaptive median filter.

Seen in the figures 7 and 9, the images using the adaptive median filter with larger max window sizes are able to be recovered much more accurately than the previous filters, creating much more cohesive pictures that preserve the edges well.

MSE	Noisy	Adaptive Median Filter (Max: 7x7)	Adaptive Median Filter (Max: 19x19)
Test1Noise2	88.97	2328.17	464.27
Test2Noise2	87.18	2726.40	962.47

Figure 11. Table of the MSE values for each test image and the given filter applied to it.

Processing Time (seconds)	Adaptive Median Filter (Max: 7x7)	Adaptive Median Filter (Max: 19x19)
Test1Noise2	2.93	3.14
Test2Noise2	33.21	35.29

Figure 12. Table of processing times taken to compute the filters on the test images.

The table in figure 11 shows similar results to figure 6, as the filter causes a much larger MSE, despite improving the actual look of the image. In this case though, the larger max window size allowed for the MSE to decrease slightly from the smaller max window size, showing that the increase in size can help with the pixel-to-pixel accuracy as well. Increasing the max window size does come at a cost though, with figure 12 showing that it does increase the processing time for the function, albeit not a super significant difference at this scale.

Discussion

Visually, it is very clear that the adaptive median filter restored the noisy images into the original images the best, despite having the worst MSE scores. The advantage that the adaptive median filter has is that it detects whether a pixel is an impulse (noise pixel), if it isn't then the

pixel doesn't change, where if it is then it is replaced by the median of the neighborhood. This is where the adaptive window comes into play, where the size of the window is increased until it is sufficiently sure whether the pixel is an impulse or not.

The distribution of finalized pixels in each window size shown in figures 8 and 10 allows for a more in depth analysis of how the adaptive window is used. As the window size increases, the amount of pixels finalized decreases, as it is less likely for a median to be considered an impulse as you increase the window size. These graphs also show the limits of increasing the window size. Since the bigger window sizes (17x17, 19x19) barely occur, using filters with larger max window sizes wouldn't make a difference as those larger sizes would never be used.

Processing time is very important in the field of computer science, as it can be a large limiting factor in certain projects. In my implementation of the adaptive median filter, I incorporated the use of integral histograms of each of the images, improving the processing speed of the function. This is done by pre-calculating the needed histograms of the image and storing them to allow for information such as the maximum, minimum, and median of neighborhoods with much better time complexity. This likely saved a lot of time since even with this implemented, the larger image took up to 35 seconds to compute, where a slower method and larger image could cause this number to be way higher. The downside with this method is that it can be memory expensive, as all the histograms have to be stored in memory to use.

Conclusion

The adaptive median filter can be a very important tool for pre-processing and restoring images. Seen from the results of the experiments, it was able to take very noisy images, and restore it quite well, preserving the majority of the details. Many functions in image processing,

such as feature extraction, histogram equalization, etc. can be very sensitive to noise in their calculations, making the application of this filter extremely important in fields where these operations are needed.

Works Cited

"OpenCV: Smoothing Images." *Docs.opencv.org*,

docs.opencv.org/4.x/d4/d13/tutorial_py_filtering.html.